Albert Wang

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Education

Stanford University — Starting Fall 2026
M.S. Electrical Engineering, Integrated Circuits | Admitted — *Stanford, CA*

Columbia University — Aug. 2024 – May 2026
B.S. Electrical Engineering | GPA: 3.67/4.0 | Dean's List 2024, 2025 — *New York, NY*

Bates College — Aug. 2021 – May 2024
B.A. Physics, Minor in Philosophy | GPA: 3.97/4.0 | Dean's List 2021–2024 — *Lewiston, ME*

Technical Skills

IC Design: CMOS transistor sizing, op-amp/filter design, small-signal analysis, biasing, stability; Cadence Virtuoso (schematic/layout/DRC/LVS/extraction), LTspice, Synopsys Design Compiler, PrimeTime
HDL & Digital: Verilog, VHDL, SystemVerilog; RTL/gate-level simulation, timing analysis, FSM design, synthesis
Lab & PCB: Oscilloscope, function generator, DMM, soldering; Altium, Autodesk Fusion 360
Programming: Python, C/C++, MATLAB, Java; signal processing, data analysis
Languages: English (native), Chinese (fluent)

Projects

Full-Custom 65nm VLSI Microprocessor — Fall 2025, *New York, NY*

- Designed complete 8-bit microprocessor schematic-to-layout in 65nm CMOS using **Cadence Virtuoso**; datapath includes ALU, shifter, 8×8 SRAM, decoder, and PLA control unit.
- Achieved 100 MHz with 88.8% timing margin, 4,100 μm^2 area; completed full DRC/LVS verification; documented parasitic characterization (2–3× delay increase) in 18-page technical report.
- Post-layout power optimization: 876 μW total consumption; analyzed process, voltage, and temperature sensitivity.

64-Tap FIR Filter with Clock Domain Crossing — Fall 2025, *New York, NY*

- Designed 64-tap 16-bit FIR filter in **Verilog** with asynchronous FIFO, MAC with saturation, FSM controller.
- Synthesized using **Synopsys Design Compiler** (IBM 130nm) achieving 100 MHz timing closure, 46,551 μm^2 area; validated via gate-level simulation with SDF back-annotation.
- Power analysis via **PrimeTime PX**: 3.384 mW total, 0.034 mW/MHz efficiency, 100% VCD annotation.

Two-Stage CMOS Op-Amp with Miller Compensation — Fall 2025, *New York, NY*

- Designed two-stage OTA with Miller compensation: gain = 4 V/V, 76 dB open-loop, 47° phase margin, 203 kHz.
- Optimized transistor sizing (W/L 4–184) for 700 μA budget with <0.1% DC accuracy; 5.4 μs, <6.5% overshoot.
- Performed AC/DC/transient analysis in **Cadence Spectre**; verified stability across process, voltage, and temperature corners.

Power Distribution PCB – Columbia Space Initiative — Fall 2024, *New York, NY*

- Led electronics team designing power distribution PCB in **Autodesk Fusion 360**; integrated overcurrent protection for high-power rocket avionics subsystems.

Experience

Hardware Security Intern (PUF/HSM) — July 2025 – September 2025
eMemory Technology Inc. — *Hsinchu, Taiwan*

- Characterized PUF-backed HSM hardware interfaces and mapped to PKCS#11 framework; co-authored API specs for partitions, protocols, HA/backup, and operations.
- Aligned interface designs with **FIPS 140-3** requirements; identified compliance checkpoints and test methodologies for certification.

Research

NSF-REU Computational Physics Researcher — May 2024 – October 2024
Clarkson University — *Potsdam, NY*

- Developed projection-based learning algorithm in **MATLAB** achieving 100× speedup for photonic crystal simulations; co-authored peer-reviewed paper at SPIE Photonics West 2025 (DOI: 10.1117/12.3028208).

Undergraduate Researcher, Neutrino Detection — May 2023 – May 2024
University of Massachusetts Amherst — *Amherst, MA*

- Investigated SiPM photon detection efficiency in liquid xenon for nEXO neutrinoless double beta decay experiment; developed waveform analysis pipeline in **Python**/Geant4; presented at APS April Meeting 2024.